

Functional Requirements

Project: Rising Water – AI-Based Flood Prediction System

Requirement ID	Description
FR-1	The system shall allow users to enter 20 flood-related environmental parameters.
FR-2	The system shall validate all user inputs before prediction.
FR-3	The system shall preprocess the input using the trained scaler.
FR-4	The system shall predict flood probability using the Machine Learning model.
FR-5	The system shall display the flood probability score.
FR-6	The system shall classify the result as Low, Medium, or High Risk.
FR-7	The system shall provide a simple web-based interface.
FR-8	The system shall allow repeated predictions without restarting the application.
FR-9	The system shall execute predictions within a few seconds.
FR-10	The system shall handle invalid inputs gracefully.

Conclusion

These functional requirements define the core features of the Rising Water Flood Prediction System.